

UF HEALTH SHANDS CORE POLICY AND PROCEDURE

POLICY NUMBER: CP02.008

CATEGORY: Patient Care

REVIEW RESPONSIBILITY/DOCUMENT OWNER: Manager, Transfusion Services

TITLE: Adult Emergency Transfusion and Massive Transfusion Protocol (MTP)

POLICY: Adult patients requiring damage control and resuscitation with blood products should have rapid initiation of blood products in an appropriate 1:1:1:1 (Plasma: Red Blood Cells: Platelets: Cryoprecipitate) ratio or Whole Blood usage. Emergency transfusion/MTP should be managed by the attending provider/designee.

PURPOSE: To define when and who may enact an Emergency transfusion/MTP and management parameters. Providers should utilize this Policy to ensure correct blood product ratio, or whole blood, and documentation is appropriate.

For Pediatric MTP: See [CP02.018 – PEDIARIC Emergency Transfusion and Massive Transfusion Protocol \(MTP\)](#)

APPROVED:

 4/1/2024 | 10:34 AM EDT

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DEFINITIONS:

- A. Emergency Release Blood Products – Any blood product, any amount less than three (3) units; Call Blood Bank 733-0900. Order and release in EPIC. May retrieve blood from direct access emergency blood refrigerator.
- B. Urgent Pack – Three (3) units Red Blood Cells and three (3) units Plasma emergently released for actively bleeding cases while determining if bleed is controlled. Call Blood Bank 733-0900. Order in EPIC. May retrieve blood from direct access emergency blood refrigerator. Escalate to MTP if appropriate.
- C. Adult Massive Transfusion- Need for the rapid transfusion of whole blood or more than ten (10) units of blood products administered in an appropriate ratio of 1:1:1:1 transfusion.
- D. Low Titer Group O Whole Blood (LTOWB)- Leukoreduced using a platelet sparing filter, is the initial product of choice for trauma MTP activations, and may also be used for emergent transfusions during trauma resuscitations not requiring MTP activation (e.g. Urgent Pack or Emergency Blood may be LTOWB pending the determination of need for MTP).

CORE PROCEDURE:

I. Massive Transfusion Protocol (MTP)

MTP should be triggered when life-threatening hemorrhage has been identified, inclusive of physiologic changes. In addition to the delivery of blood products, other goals during an MTP are:

- A. Prevent dilutional coagulopathy
- B. Avoid hypothermia: Maintain core temperature > 35° C
- C. Monitor appropriate laboratory tests and physiologic parameters
- D. Avoid wastage of blood products via coordination and communication between the Blood Bank and identified individuals at the treatment site regarding the status of these patients
- E. Maintain INR ≤1.7
- F. Maintain fibrinogen > 150-200 mg/dL with active bleed
- G. Maintain platelet count > 50,000/microliter while patient is actively hemorrhaging or > 100,000 for central nervous system bleeding, CNS surgery or Ophthalmic surgery
- H. Prevent symptomatic anemia
- I. Prevent hyperkalemia
- J. Prevent hypocalcaemia
- K. Avoid transfusion reactions

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II. Materials Required in an MTP

- A. Documentation of MTP is entirely in the electronic medical record. During downtime, initiate documentation on the MTP Order Form ([PS66259](#)).
- B. Satellite Blood Bank refrigerators
 - 1. Located in South Tower Emergency Department, South Tower Operating Room, North Tower Operating Room 2352, North Tower PCICU 10-2, and East (HVN) Tower CICU 7711.
 - 2. Blood Bank personnel are responsible for regulating the inventory and stocking of the satellite refrigerators.
 - 3. Blood Bank satellite refrigerators must be maintained and operated in accordance with the standards and regulations of the Association for the Advancement of Blood and Biotherapies (AABB), The Joint Commission, and the FDA.
- C. Product transport coolers
- D. Product coolant packs
- E. Pre-warmed crystalloids
- F. Large bore intravenous (IV) catheters, preferably 16-gauge IV or #9 French central line introducer
- G. Rapid infusion warming device. Must have one or more of the following:
 - 1. Fluid Management System (FMS) i.e. Belmont or equivalent
 - 2. Level I Fluid Warmer

Note: Do not use an electric blanket to warm the MTP products (e.g. bear huggers)

- H. Air convection warming system (e.g. bear hugger) to maintain core temperature
- I. Point of Care (POC) laboratory value measurements

III. Specimen Requirements

- A. A new specimen must be sent to the Blood Bank as soon as an MTP is initiated if patient does not have current Type and Screen.
- B. The specimen must be legibly labeled with:
 - 1. Patient name (first and last name or trauma designated name which must not be changed until after the MTP is completed, see section III, E);
 - 2. Patient Medical Record Number (if available);

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3. Date and time of collection;
 4. Employee ID number or initials of who collected the specimen; and
 5. Verification form.
- C. Preferred specimen is venous blood collected in 6 mL K₂ EDTA (pink/purple top tube).
- D. The patient specimen will be rejected, if not labeled properly, and another specimen requested.
- E. The Hospital requires that there be two (2) ABO/RH specimens, drawn different times, on file prior to type specific blood products being released. The Blood Bank will notify staff if a second blood specimen needs to be drawn.
- F. If a sample is not delivered to the Blood Bank, the Blood Bank cannot provide compatible blood products.**

Note: Refer to Policy CP 2.006- Changing Trauma Name to Legal Name for proper name in the alignment with ongoing care needs.

IV. Procedure for Initiating MTP

- A. Only the following individuals may initiate an MTP in the Emergency Department, Operating Room, Intensive Care Unit and Labor and Delivery:
1. Attending physician
 2. Fellow
 3. Chief Resident/Acting Chief Resident
- B. Use of Whole Blood may only be ordered by the Trauma Team Attending/Designee, and is limited to six (6) units of Whole Blood per incident.
- C. The responsible physician will give a verbal order and must state "Initiate Massive Transfusion Protocol" to the designated nurse communicator.
- D. The MTP initiation start order must be documented in the medical record.
- E. A nurse will be designated in each treatment area as NURSE COMMUNICATOR who will serve as the central person in communication with the Blood Bank in that treatment area. The role of the nurse communicator is paramount to a successful MTP and patient outcome.
1. When patients move between treatment areas, it is the responsibility of the nurse communicator to interact directly with his or her counterpart in the new treatment area to assure continuity of the process. The nurse communicators are the ED coordinator, charge nurse, or RN designee.
 2. Immediately call the Blood Bank on a designated phone via speed dial or designated Blood Bank (Extension 2-7320 or 733-0900).

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3. Inform the Blood Bank of the following:
 - a. An MTP has been initiated
 - b. Patient Name, MRN, Gender, Age, Weight, Ordering physician, Location and phone Number
4. Product turn-around-time, from order to implementation, should not take longer than twenty (20) minutes.
5. An overview of the MTP process can be found in the MTP algorithm ([Appendix A](#)).

F. Dispensing Blood Products:

1. Blood products, including whole blood, will be dispensed per current practices.
2. Blood products will be dispensed in the approved Blood Bank coolers.
3. If blood ratio-driven MTP therapies are used in place of whole blood, appropriate ratios of 1:1:1:1 are shipped in predetermined packages as outline in the MTP Adult Schedule ([Appendix B](#)).
4. Note: Shipped ratios of ratio-driven MTP is not intended to be a mandate for transfusion of those products but to ensure adequate products are available at the bedside. These ratios were determined to be reasonable and effective by a team of hematologists and transfusion medicine specialists; however, the responsible physician can decide which products to transfuse based on the patient's clinical and laboratory value status.
5. If the clinician determines that a different ratio of products is needed, the clinician must communicate the change and whether the blood products are in place or, or in addition to, the next required shipment.

G. A DESIGNATED RUNNER will transport products and specimens.

1. Products will be transported in designated blood transport coolers.
2. Runner is responsible to return blood coolers to the Blood Bank after the MTP is terminated.
3. Runner is responsible to bring specimens to the Blood Bank and blood products to the treatment area during the MTP. Specimens must meet labeling criteria set forth in Section III-A.
4. The runner will sign the MTP Pack # slip, documenting the date, time, and person picking up the blood products.

V. Continuance/Changes/Discontinuance of the Protocol

A. Continuance of the MTP

1. MTP may be continued into another treatment area.

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2. The nurse communicator from the originating site must perform the following:
 - a. Notify the Blood Bank
 - b. Transport available products in a cooler with the patient
 - c. Have face-to-face hand-off with the nurse communicator from the next location
 - d. Continue complete documentation of MTP in the EMR.
 - i. Utilize paper MTP forms during downtime

B. Changes in MTP

1. If there is a change in the need for blood products during the protocol the responsible physician must submit the change via a verbal order to the nurse communicator to avoid any confusion and deviation from the protocol.
2. This order must describe the change and the duration of the change.

C. Discontinuance of MTP

1. The responsible physician must notify the nurse communicator that MTP is to be discontinued.
2. The nurse communicator must call the Blood Bank immediately via the designated phone/ (Extension 2-7320 or 733-0900).
3. The Blood Bank will automatically terminate the MTP if there is a one (1) hour period without blood product pick up from the time of the last shipment.

VI. Core MTP Procedure

- A. MTP initiation should be followed in the steps above.
- B. Blood Bank will restock the Trauma Blood Bank refrigerators upon completion of the MTP. If more units are needed prior to restocking, runners will go to Blood Bank to pick up additional MTP shipments.
- C. Platelets and Cryoprecipitate are sent from the main Blood Bank (ST G110).
- D. Whole Blood is located in the top Blood Bank refrigerator in the Trauma Bay area, and should be used only when ordered by the Trauma Team Attending/Designee. Whole Blood (LTOWB) may be transfused during trauma resuscitation efforts regardless of whether or not an MTP has been activated. A limited supply of Whole Blood is also available from the main South Tower Blood Bank (G110).
- E. Plasma and RBC units are stored in the bottom Blood Bank refrigerator in the Trauma Bay area, and in the South Tower OR, North Tower OR, and HVN CICU (unit 77).

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Note: The same red kiosk in the Trauma Bay controls both the top and bottom blood refrigerators in the Trauma Bay area.

F. Ratio-Driven Transfusion Procedure

1. The Blood Bank will place six (6) units of pRBC and six (6) units of plasma along with coolant in an approved blood transport cooler.
 - a. One (1) unit of platelets and one (1) pool of Cryoprecipitate (Cryo) or Intercept Fibrinogen Complex (IFC) will be transported at room temperature with every other shipment.
 - b. All blood products will be prepared according to the established Blood Bank policy.
 - c. The Blood Bank will notify the treatment area any time there is a problem or delay with a future shipment.
 - d. The Blood Bank will stay one (1) shipment ahead until the MTP is terminated.
 - e. Blood type in subsequent shipments may vary depending on type, screen, and available inventory.
 - f. Deviation from this schedule must be presented in the form of a written or electronic order

G. Tranexamic Acid (TXA)

1. Use of TXA is recommended in the adult trauma patient and the initial dose should be administered within three (3) hours from the time the injury occurred.
2. TXA should be administered per physician order and only when clinical indicators warrant intervention. TXA must be ordered through the pharmacy if the decision is made to administer.

Note: TXA will not be automatically sent with the initiation of MTP.

VII. Recommended Laboratory Monitoring

- A. Lab tests may be helpful in the treatment of the hemorrhaging patient and should be used at the discretion of the responsible physician and can include:
1. Hematocrit
 2. Platelet Count
 3. Ionized Calcium (Ca⁺)
 4. Arterial Blood Gas (ABG)
 5. PT/INR

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6. Rapid TEG (Available in the Stat Lab)

B. The following lab tests are available in the Main Lab and can be used at the discretion of the responsible MD to guide management; recommended every sixty (60) to ninety (90) minutes during massive transfusion:

1. Hemoglobin and hematocrit
2. Platelets
3. PT/INR
4. Fibrinogen
5. Blood gas with electrolytes

C. Recommended transfusion goals can include:

1. Hgb > 7-8 g/dL
2. Platelets > 50,000-100,000 (for CNS trauma, CNS/Ophthalmic surgery > 100,000)
3. INR $\leq$ 1.7
4. Fibrinogen > 150 mg/dL

D. Reasonable transfusion volumes and expected lab changes may include:

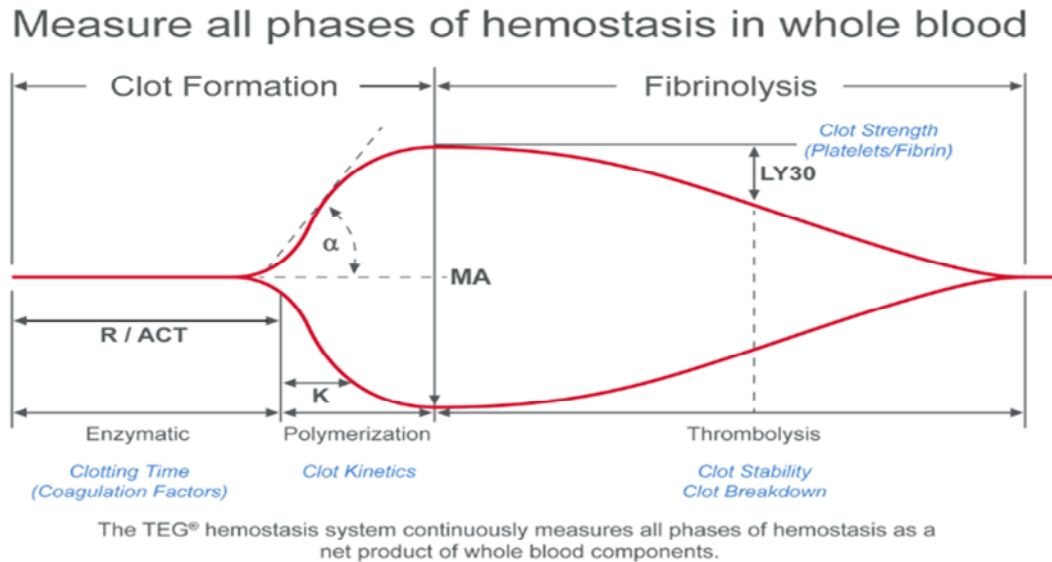
1. For RBCs, 1 unit → Hgb increased by 1 g/dL
2. For plasma, at least 15ml/kg → factor activity increased by 20%
3. For platelets, 1 Unit apheresis → platelets increased by 50,000
4. For cryoprecipitate or IFC, 5/6 pack pools → fibrinogen increase by 60-100 mg/dL

VIII. Transfusion Guided by Viscoelastic Hemostatic Assays (Thromboelastography / TEG)

- A. For clinicians comfortable with the interpretation of viscoelastic hemostatic assays, this may be the most rapid and accurate method of assessing coagulation status during MTP.
- B. Recommend standard TEG 5000 (Run by UF Stat Lab).
- C. TEG curve development can be viewed in real-time by the clinician. A brief and simplistic primer on the use of TEG 5000 follows:

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D. TEG can be interpreted in four (4) phases:

1. Clot initiation time – represented by R-time (normal value 5-10 min)
2. Rate of clot strengthening – represented by K-time (1-3 min) and alpha-angle (53-72 degrees)
3. Maximum clot strength – represented by MA (50-70 mm)
4. Fibrinolysis – represented by LY30 (0-8%)

E. The following are reasonable interpretations of TEG abnormalities:

1. Prolonged R time – factor deficiency = transfuse plasma
2. Prolonged K-time or low alpha-angle – fibrinogen deficiency = order and transfuse cryoprecipitate
3. Low MA = transfuse platelets if platelet count is <50,000 and transfuse cryoprecipitate if platelet count is ≥50,000
4. Increased percentage fibrinolysis = consider administration of an antifibrinolytic (TXA, for example, 1 gm initial infusion, then 1gm q 1hr, if clinically appropriate)

IX. Ongoing Quality Management

- A. Blood Bank will complete “MTP Report” for all MTP activations. The report will document blood products prepared and transfused for the case, and any performance issues encountered during the event.
- B. MTP activation and blood product utilization for each MTP report will be reviewed by the Blood Bank Medical Director for appropriateness.

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- C. MTP report findings will be communicated with Trauma Service for continuing performance improvement.

X. Anticipated Complications and Management

- A. **Volume overload** – The estimated blood volume for an adult is 70 mL/kg; during pregnancy, blood volume increases to 80 mL/kg. Volume overload may occur if transfusion exceeds ongoing losses, increased risk with renal or cardiac failure. Again, transfusion from MTP packs should always follow clinical judgment in an attempt to match ongoing blood losses or respond to specific laboratory deficiencies. The amount of blood in a pack is a logistical solution, allowing for the rapid transport of reasonable blood products to the site of clinical care, NOT a suggestion that the entire MTP pack be transfused.
- B. **Hypothermia** – Patients in the setting of traumatic hemorrhage are especially susceptible to hypothermia, which (if allowed to persist) may depress cardiac function, decrease drug metabolism, and further impair the coagulation cascade. PRBCs and FFP should be administered utilizing rapid infusion warming devices, if available, and the monitoring of temperature and use of patient warming devices is always recommended in the setting of massive hemorrhage and transfusion.
- C. **Hyperkalemia** – The potassium level of PRBC supernatant increases with storage, and massive transfusion may lead to hyperkalemia, especially in the setting of shock and acidosis. Plasma potassium should be intermittently checked, and if appropriate, therapeutic measures to treat hyperkalemia should be started by the treating physician.
- D. **Hypocalcemia** – Blood components contain citrate, which binds ionized calcium. Rapid transfusion that exceeds the liver's ability to metabolize citrate may lead to hypocalcemia. Laboratory evidence of acute low ionized calcium in the setting hemodynamic instability should prompt the initiation of therapeutic measures by the treating physician. Repeated or increased doses of supplemental calcium may be needed, especially during ongoing periods of rapid transfusion or if ionized calcium is found to be severely decreased.
- E. **Acid/base disturbances** – Many patients with ongoing hemorrhage will develop metabolic acidosis, which may depress cardiac function, decrease responsiveness to vasoactive medications, further impair the coagulation cascade, and exacerbate hyperkalemia. It should be the focus of the treating physician to identify and treat the underlying causes of acid/base disturbances.
- F. **Transfusion reactions** – The risk of a transfusion reaction increases with high volume transfusions. Standard procedures for checking blood products prior to administration should be followed, and a high index of suspicion for transfusion reaction should be maintained during administration.

XI. Quality Improvement Process Following MTP

- A. Every MTP will be subject to review by the Transfusion Committee and the specific department committee involved.

ASSOCIATED POLICIES:

CP02.006 – Changing Trauma Name to Legal Name

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KEY WORDS: Trauma, MTP, blood administration, TEG, TXA, blood bank